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RESEARCH ARTICLE

Two Independent Single-Loop Voltage Mode Control Method for 3-Level Buck Converter

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ABSTRACT Compared to a basic buck converter, the 3-level buck converter operates with the output inductor at twice the switching frequency. Additionally, the voltage across its flying capacitor contributes to reducing the slope of the inductor current. These factors can result in a reduction in the maximum output current ripple by up to four times, enabling the use of a smaller output inductor. Moreover, the voltage stress on the switches is only half of the input voltage. Therefore, the 3-level buck converter is advantageous for achieving high efficiency and high power density. To maintain these advantages, the voltage across the flying capacitor must be kept at half of the input voltage. Conventionally, this condition is usually ensured by using peak or valley current mode control methods. However, these methods have complex control circuits or algorithms and are vulnerable to noise interference because their operation is based on instantaneous peak or valley current sensing. Additionally, different control configurations must be employed for $M < 0.5$ and $M > 0.5$, where M is the input-to-output voltage conversion ratio. In this paper, we propose a two independent single-loop voltage mode control method that can guarantee the flying capacitor voltage remains at half of the input voltage. The proposed control method is simple to implement and does not require current sensing or complex control algorithms, allowing the use of simple and low-cost digital or analog controllers. The method is also robust to noise interference because it only operates based on the output and flying capacitor voltage. Furthermore, this method ensures normal operation across all values of M without changing the control configuration. The experimental results from a 500 W-rated prototype confirmed the validity of the proposed control method.

INDEX TERMS 3-level buck converter, high efficiency, high power density, slim converter, voltage mode control.

NOMENCLATURE

f_{sw} :	Switching frequency.
T_s :	Switching period.
D_i :	Duty cycle of the main switch Q_{HSi} ($i = A, B$).
V_{SW} :	Left side node voltage of the output inductor L_o .
i_{Lo} :	Current of the output inductor L_o .
i_{CFly} :	Current of the flying capacitor C_{Fly} .
R_{Load} :	Load resistance.
M :	Input voltage V_{in} to output voltage V_o conversion ratio.

M_F :	V_{in} to flying capacitor voltage V_{Fly} conversion ratio.
V_{Fly_ero} :	Control signal of the V_{Fly} controller.
V_{o_ero} :	Control signal of the V_o controller.
r_o :	Equivalent series resistance of the output capacitor.
$G_{cf}(s)$:	Transfer function of the V_{Fly} control compensator.
$G_{cv}(s)$:	Transfer function of the V_o control compensator.
$G_{da}(s)$:	Transfer function of the differential amplifier.
$G_d(s)$:	Transfer function of the computation delay and zero-order holder, including modulation gain.

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- $G_{oo}(s)$: Transfer function of the output voltage control signal perturbation Δv_{o_ero} to output voltage perturbation Δv_o .
- $G_{ff}(s)$: Transfer function of the flying capacitor voltage control signal perturbation Δv_{Fly_ero} to the flying capacitor voltage perturbation Δv_{Fly} .
- SC: Switched Capacitor.
- LS: Low Side.
- 2PSC: 2-Phase Series Capacitor.
- P-CMC: Peak Current Mode Control.
- V-CMC: Valley Current Mode Control.
- PWM: Pulse Width Modulation.

I. INTRODUCTION

The current trend toward large-screen, ultra-slim TVs has increased demand for slim and compact circuit board designs, including display panels. However, reducing the size of power circuits is an extremely challenging task because it involves optimizing components such as inductors and capacitors, which consume a substantial amount of space. Achieving ultra-slim designs for non-isolated dc-dc converters that are used as post-regulators for LED drivers and mainboards in TVs is particularly challenging. This difficulty arises primarily due to the essential high-current power inductors in the power supplies [1], [2].

Among the recognized non-isolated dc-dc converters, the 3-level switched capacitor (SC) buck converter depicted in Fig. 1(a) only comprises switches and capacitors, providing substantial benefits for the requirement of a slim design. Additionally, because this configuration operates based on the charge pump of capacitors, it features high efficiency, simple operation, and ensures that the voltage stress on all switches remains at half the input voltage [3], [4], [5], [6]. The output voltage is minimally influenced by the pulse width or operating frequency and remains constant at approximately half of the input voltage. Additionally, it cannot effectively regulate the output voltage in response to load variations and input voltage fluctuations.

The 3-level buck converter displayed in Fig. 1(b) features a circuit configuration with an output inductor for a 3-level SC buck converter, and its output voltage can be controlled as $V_o = DV_{in}$ by adjusting the duty cycle D of Q_{HSA} and Q_{HSB} . Furthermore, because the output inductor operates at twice the switching frequency, it can reduce the maximum inductor current ripple and the maximum output voltage ripple by up to four and eight times, respectively, compared to a basic buck converter. This renders the 3-level buck converter highly advantageous for achieving high power density. Moreover, since the voltage stress on the switch can be guaranteed to be $V_{in}/2$, this results in lower switching losses and higher efficiency [7], [8], [9], [10], [11].

The 3-level low side (LS) buck converter displayed in Fig. 1(c) functions similarly to a 3-level SC buck converter (with the exception of L_o and C_o) and maintains the voltage across C_L at approximately $V_{in}/2$. Moreover, the circuit, composed of Q_{LSA} , Q_{LSB} , and L_o , functions as a synchronous buck

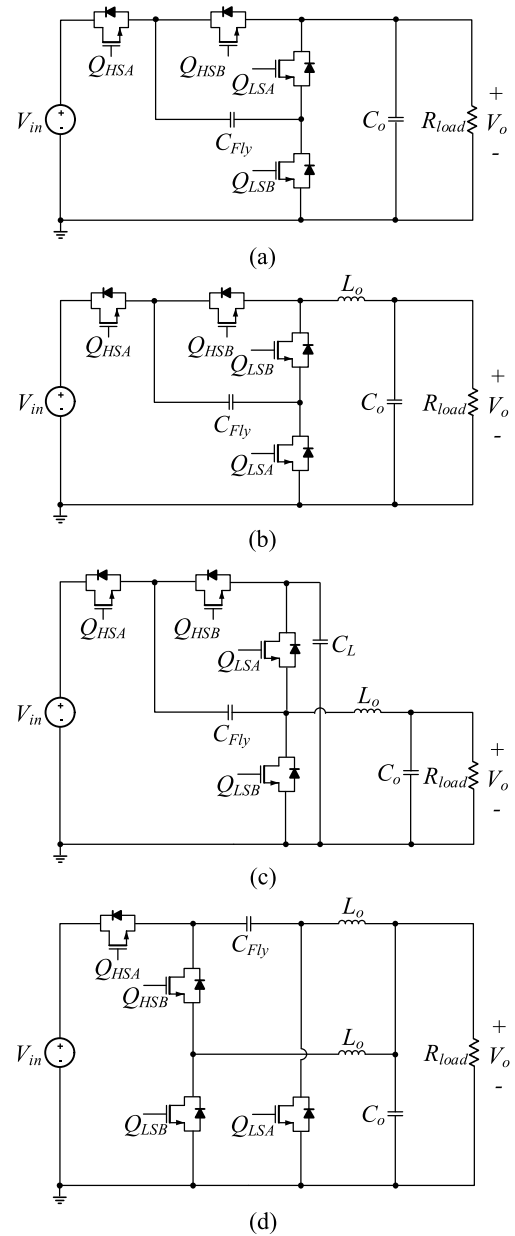


FIGURE 1. Conventional buck converters for achieving high efficiency and high power density: (a) 3-level SC buck converter, (b) 3-level buck converter, (c) 3-level LS buck converter, and (d) 2PSC buck converter.

converter, using the C_L voltage as the input and generating the C_o voltage as the output. Therefore, the output voltage is maintained at $V_o = 0.5DV_{in}$, which is determined by the duty cycle D of Q_{LSA} with the input-to-output voltage conversion ratio ($M = V_o/V_{in}$), which is restricted to a maximum of 0.5. The maximum inductor current ripple and output voltage ripple can be reduced by half compared to that of a basic buck converter, resulting in smaller values for L_o and C_o . Additionally, this circuit is advantageous in terms of reducing switching losses by ensuring that the switch voltage stress is $V_{in}/2$ [12], [13].

The 2-phase series capacitor (2PSC) buck converter displayed in Fig. 1(d) requires two inductors operating with a

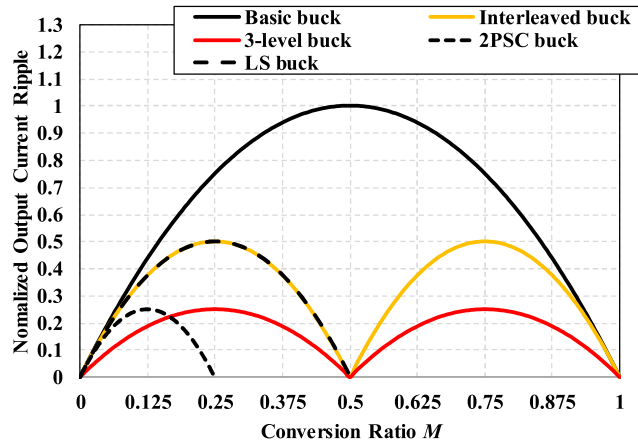


FIGURE 2. Comparison of normalized output current ripple for each buck converter.

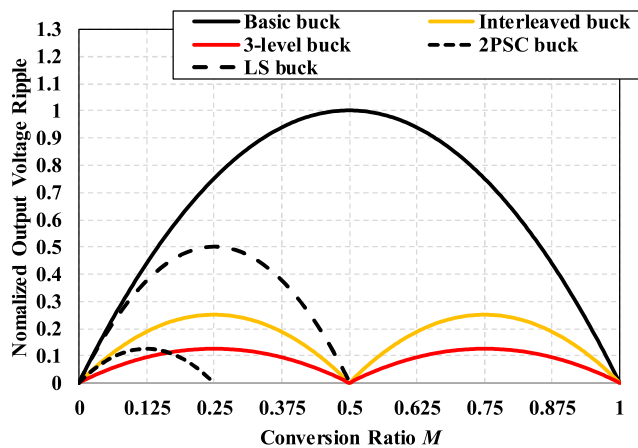


FIGURE 3. Comparison of normalized output voltage ripple for each buck converter.

180° phase difference. The output voltage is equivalent to that of a 3-level LS buck converter and is determined by $V_o = 0.5DV_{in}$ using the duty cycle D of Q_{HSA} and Q_{HSB} . However, for correct operation, it is crucial to maintain the duty cycle below 0.5, which restricts the voltage conversion ratio M to a maximum of 0.25. The maximum inductor current ripple can be reduced by up to two times compared to a basic buck converter. Additionally, its interleaved operation can reduce the maximum output current ripple by up to four times compared to a basic buck converter. The maximum output voltage ripple can also be reduced by up to eight times, equivalent to that of a 3-level buck converter. This converter maintains the switch voltage stress at $V_{in}/2$, which is advantageous in terms of reducing switching losses [14], [15], [16], [17].

Fig. 2 and Fig. 3 display comparisons of the normalized output current ripple and voltage ripple for each converter, respectively, including a basic buck converter and an interleaved buck converter. Table 1 outlines the main characteristics of conventional buck converters, where the 3-level and 2PSC buck converters exhibit the best performance in terms of output current and voltage ripple. However, it should be noted that the 2PSC buck converter requires two inductors, each with a maximum current ripple that is twice that of

TABLE 1. Comparison of conventional buck converters.

Feature	Conventional Buck Converters				
	Basic	Interleaved	3-level	LS	2PSC
Number of Inductors	1	2	1	1	2
Max. M	1	1	1	0.5	0.25
Max. Norm. Output current ripple	1	0.5	0.25	0.5	0.25
Max. Norm. Output voltage ripple	1	0.25	0.125	0.5	0.125
Switch Voltage Stress	V_{in}		$0.5V_{in}$		

a 3-level buck converter. Additionally, the 2PSC buck converter's maximum voltage conversion ratio is limited to 0.25, whereas the 3-level buck converter is superior in terms of securing a wide input-to-output operating range. Conversely, the 3-level buck converter requires an additional means of control for the flying capacitor voltage. This is because, it only achieves optimum performance (in terms of output current and voltage ripple) when the flying capacitor voltage is maintained at half of the input voltage. Failure to maintain this condition can result in increased current and voltage ripple, as well as circuit losses.

Conventional control methods, such as peak and valley current mode control (CMC), have been proposed to ensure a balanced inductor current and maintain the flying capacitor voltage at half of the input voltage. However, as detailed in the manuscript, these methods have complex control circuits (or algorithms) and are vulnerable to noise interference because their operation is based on instantaneous peak or valley current sensing. Additionally, due to current subharmonic oscillations, these methods require slope compensation as an additional means of ensuring normal operation in all input and output voltage operating ranges [18], [19], [20]. Furthermore, different control configurations must be employed for $M < 0.5$ and $M > 0.5$, where M is the input-to-output voltage conversion ratio.

To overcome these limitations of the conventional CMC methods, we propose a two independent single-loop voltage mode control method in this paper that guarantees the flying capacitor voltage remains at half of the input voltage across all M values, while simultaneously controlling the output voltage. The proposed control method is simple to implement and does not require current sensing or complex control algorithms. It is also robust to noise interference because it only operates based on the output and flying capacitor voltage. Furthermore, the proposed control method ensures normal operation across all M values without changing the control configuration. This paper builds on our previous work [21] by providing a more detailed explanation. In Section II, we discuss the voltage of the flying capacitor and the current of the inductor based on the operating principle of the 3-level buck converter, addressing these aspects in greater detail compared

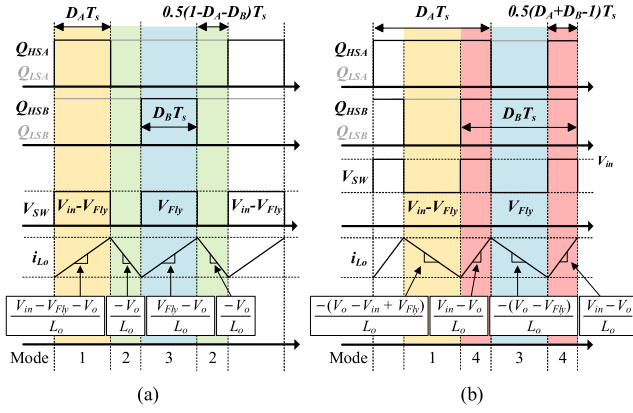


FIGURE 4. Key operation waveforms of a 3-level buck converter: (a) $M < 0.5$, (b) $M > 0.5$.

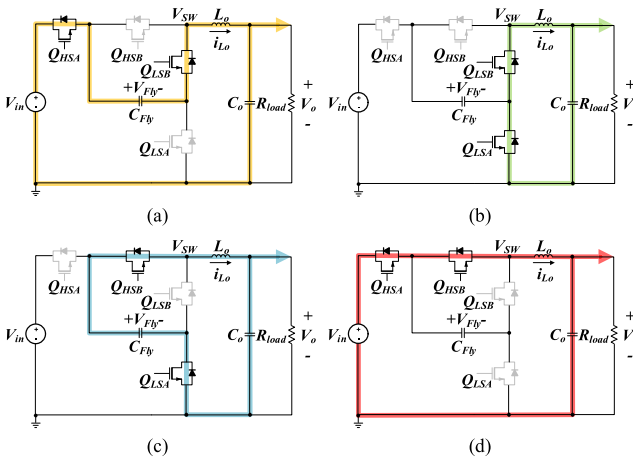


FIGURE 5. Four operating modes of a 3-level buck converter: (a) Mode 1, (b) Mode 2, (c) Mode 3, and (d) Mode 4.

to [21]. Section III covers the conventional and proposed control methods for ensuring the voltage of the flying capacitor remains at half of the input voltage, including a more detailed analysis and modeling of the proposed control method. The experimental results from a 500 W-rated prototype to validate the proposed control method are presented in Section IV, and the paper is concluded in Section V.

II. 3-LEVEL BUCK CONVERTER

A. OPERATION PRINCIPLE

A 3-level buck converter comprises two main switches (Q_{HSA} and Q_{HSB}) and two synchronous rectifiers (Q_{LSA} and Q_{LSB}). Q_{HSA} and Q_{LSA} in addition to Q_{HSB} and Q_{LSB} form separate switching groups, each driven with a 180° phase difference and operating complementarily, as depicted in Fig. 4. The converter operates in two regions based on the voltage conversion ratio (M): a low-gain region ($M < 0.5$) and a high-gain region ($M > 0.5$), as depicted in Fig. 4(a) and (b). Each region comprises four modes, as illustrated in Fig. 5.

Mode 1: In Fig. 5(a), Q_{HSA} and Q_{LSB} are on, resulting in $V_{SW} = V_{in} - V_{Fly}$. The voltages across Q_{HSB} and Q_{LSA} are V_{Fly} and $V_{in} - V_{Fly}$, respectively, and the voltage across the inductor is $V_{in} - V_{Fly} - V_o$. Therefore, the slope of the

inductor current is $(V_{in} - V_{Fly} - V_o)/L_o$, as depicted in Fig. 4(a) and (b). During this period, the flying capacitor is charged by the inductor current.

Mode 2: In Fig. 5(b), Q_{LSA} and Q_{LSB} are on, resulting in $V_{SW} = 0$ V. The voltages across Q_{HSA} and Q_{HSB} are $V_{in} - V_{Fly}$ and V_{Fly} , respectively, and the voltage across the inductor is $-V_o$. Therefore, the slope of the inductor current is $-V_o/L_o$, as depicted in Fig. 4(a).

Mode 3: In Fig. 5(c), Q_{HSB} and Q_{LSA} are on, resulting in $V_{SW} = V_{Fly}$. The voltages across Q_{HSA} and Q_{LSB} are $V_{in} - V_{Fly}$ and V_{Fly} , respectively, and the voltage across the inductor is $V_{Fly} - V_o$. Therefore, the slope of the inductor current is $(V_{Fly} - V_o)/L_o$, as depicted in Fig. 4(a) and (b). During this period, the flying capacitor is discharged by the inductor current.

Mode 4: In Fig. 5(d), Q_{HSA} and Q_{HSB} are on, resulting in $V_{SW} = V_{in}$. The voltages across Q_{LSA} and Q_{LSB} are $V_{in} - V_{Fly}$ and V_{Fly} , respectively, and the voltage across the inductor is $V_{in} - V_o$. Therefore, the slope of the inductor current is $(V_{in} - V_o)/L_o$, as depicted in Fig. 4(b).

In the low-gain region, as depicted in Fig. 4(a), the converter operates in the sequence mode 1 $\rightarrow$ mode 2 $\rightarrow$ mode 3 $\rightarrow$ mode 2, while it operates in the sequence mode 1 $\rightarrow$ mode 4 $\rightarrow$ mode 3 $\rightarrow$ mode 4 in the high-gain region, as depicted in Fig. 4(b).

B. INDUCTOR CURRENT BALANCING CONDITION

In the low-gain region, the falling slope of the inductor current is determined by V_o , as depicted in mode 2. In contrast, the rising slope is determined by $V_{in} - V_{Fly} - V_o$ or $V_{Fly} - V_o$, as depicted in modes 1 and 3. Similarly, the rising slope of the inductor current is determined by $V_{in} - V_o$ in the high-gain region, as depicted in mode 4. In contrast, the falling slope is determined by $-V_{in} + V_{Fly} + V_o$ or $-V_{Fly} + V_o$, as displayed in modes 1 and 3. Therefore, to ensure the current balance of the inductor for every switching half-cycle, the slope of the inductor current should remain the same in the operating modes displayed in modes 1 and 3. Specifically, in the low-gain region, $V_{in} - V_{Fly} - V_o$ should equal $V_{Fly} - V_o$, whereas $-V_{in} + V_{Fly} + V_o$ should equal $-V_{Fly} + V_o$ in the high-gain region. Therefore, maintaining V_{Fly} at $V_{in}/2$ ensures the current balance of the inductor for every switching half-cycle, as illustrated in Fig. 4. In both the low- and high gain regions, the volt-second balance of the inductor must be satisfied at the steady state to ensure current balance in the inductor for every switching half-cycle. In the low-gain region, the volt-second balance of the inductor can be expressed as follows:

$$D_A (V_{in} - V_{Fly} - V_o) = D_B (V_{Fly} - V_o) \\ = 0.5V_o (1-D_A - D_B). \quad (1)$$

In the high-gain region, the volt-second balance of the inductor can be expressed as follows:

$$(1-D_A) (V_o - V_{Fly}) = (1-D_B) (V_o - V_{in} + V_{Fly}) \\ = 0.5 (V_{in} - V_o) (D_A + D_B - 1). \quad (2)$$

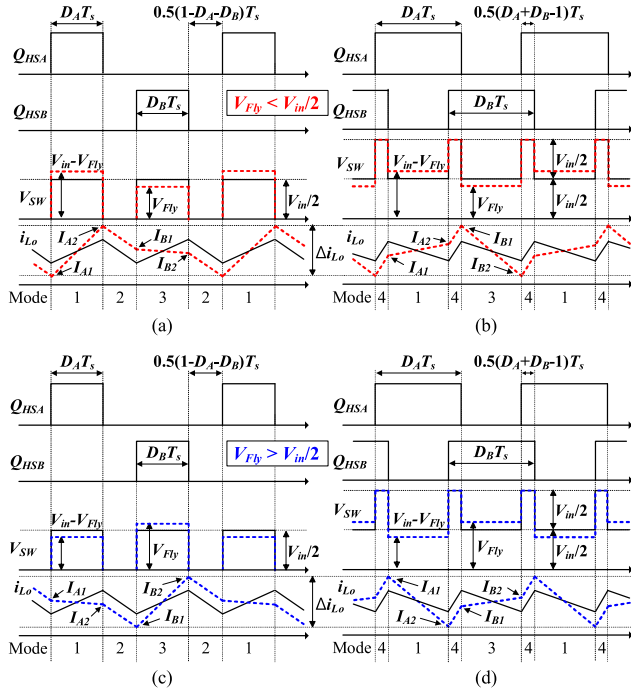


FIGURE 6. Relationship between V_{Fly} and inductor current: (a) $M < 0.5$, $V_{Fly} < V_{in}/2$, (b) $M > 0.5$, $V_{Fly} < V_{in}/2$, (c) $M < 0.5$, $V_{Fly} > V_{in}/2$, and (d) $M > 0.5$, $V_{Fly} > V_{in}/2$.

As discussed previously, when $V_{Fly} = V_{in}/2$ is ensured from both (1) and (2), $D_A = D_B$ can be satisfied. Furthermore, the voltage conversion ratio $M = D$ (where $D = D_A = D_B$) can be achieved, meaning the voltage conversion ratio of the 3-level buck converter is the same as that of the basic buck converter. Consequently, when $V_{Fly} = V_{in}/2$ is ensured, all switch voltage stresses are balanced at $V_{in}/2$, and the inductor current balance for every switching half-cycle is guaranteed.

C. RELATIONSHIP BETWEEN INDUCTOR CURRENT RIPPLE AND FLYING CAPACITOR VOLTAGE

As illustrated in Fig. 6, an imbalance in the inductor current occurs if V_{Fly} is not maintained at $V_{in}/2$, resulting in increased current stress and circuit losses due to the higher current ripple.

When V_{Fly} decreases, the slope of the inductor current in mode 1 increases, while the slope of the inductor current in mode 3 decreases. Conversely, when V_{Fly} increases, the slope of the inductor current in mode 1 decreases, while the slope of the inductor current in mode 3 increases. Accordingly, in the low-gain region, the inductor current ripple in mode 1 is larger when $V_{Fly} < V_{in}/2$ than when $V_{Fly} = V_{in}/2$ (black line in Fig. 6(a)), as indicated by the red dotted line in Fig. 6(a). Similarly, the inductor current ripple in mode 3 is larger when $V_{Fly} > V_{in}/2$ than when $V_{Fly} = V_{in}/2$ (black line in Fig. 6(c)), as indicated by the blue dotted line in Fig. 6(c). In the high-gain region, the inductor current ripple in mode 3 is larger when $V_{Fly} < V_{in}/2$ than when $V_{Fly} = V_{in}/2$ (black line in Fig. 6(b)), as indicated by the red dotted line in Fig. 6(b). Similarly, the inductor current ripple in mode 1 is larger when

$V_{Fly} > V_{in}/2$ than when $V_{Fly} = V_{in}/2$ (black line in Fig. 6(d)), as indicated by the blue dotted line in Fig. 6(d).

Therefore, in the low-gain region, the inductor current ripple Δi_{Lo} of a 3-level buck converter according to M_F is expressed as follows:

$$\Delta i_{Lo, M_F < 0.5} = \frac{V_{in}}{L_o f_{sw}} D_A (1 - M_F - M), \quad (3)$$

$$\Delta i_{Lo, M_F > 0.5} = \frac{V_{in}}{L_o f_{sw}} D_B (M_F - M), \quad (4)$$

where M_F represents the V_{in} to V_{Fly} conversion ratio, and f_{sw} is the switching frequency. In the high-gain region, Δi_{Lo} of a 3-level buck converter is given by

$$\Delta i_{Lo, M_F < 0.5} = \frac{V_{in}}{L_o f_{sw}} (1 - D_A) (M - M_F), \quad (5)$$

$$\Delta i_{Lo, M_F > 0.5} = \frac{V_{in}}{L_o f_{sw}} (1 - D_B) (M - 1 + M_F). \quad (6)$$

The current-second balance of the flying capacitor C_{Fly} must be satisfied at a steady state in both the low- and high-gain regions. In the low-gain region, C_{Fly} is charged during the duration $D_A T_s$ of the operation in mode 1 and discharged during the duration $D_B T_s$ of the operation in mode 3. Therefore, based on Fig. 6(a) and (c), the following is satisfied:

$$(I_{A1} + I_{A2}) D_A T_s = (I_{B1} + I_{B2}) D_B T_s, \quad (7)$$

where I_{A1} and I_{A2} are the start and end currents of the inductor in mode 1, respectively, while I_{B1} and I_{B2} are the start and end currents of the inductor in mode 3. In the high-gain region, C_{Fly} is charged during the duration $(1 - D_B) T_s$ of the operation in mode 1 and discharged during the duration $(1 - D_A) T_s$ of the operation in mode 3. Therefore, based on Fig. 6(b) and (d), the following is satisfied:

$$(I_{A1} + I_{A2}) (1 - D_B) T_s = (I_{B1} + I_{B2}) (1 - D_A) T_s. \quad (8)$$

Additionally, in the low-gain region, the slope of the inductor current during the duration $0.5(1 - D_A - D_B) T_s$ of the operation in mode 2 is determined by V_o . Thus, based on Fig. 6(a) and (c), the following is satisfied:

$$I_{A2} - I_{B1} = I_{B2} - I_{A1}. \quad (9)$$

In the high-gain region, the slope of the inductor current during the duration $0.5(D_A + D_B - 1) T_s$ of the operation in mode 4 is determined by $V_{in} - V_o$. Thus, based on Fig. 6(b) and (d), the following is satisfied:

$$I_{A1} - I_{B2} = I_{B1} - I_{A2}. \quad (10)$$

From (9) and (10), $I_{A1} + I_{A2} = I_{B1} + I_{B2}$ is satisfied. Therefore, from (7) and (8), the following can be obtained:

$$D_A = D_B. \quad (11)$$

Therefore, the duty cycles D_A and D_B are the same regardless of V_{Fly} . In both the low- and high-gain regions, the volt-second balance of the inductor must be satisfied at the steady state for every switching cycle. In the low-gain region,

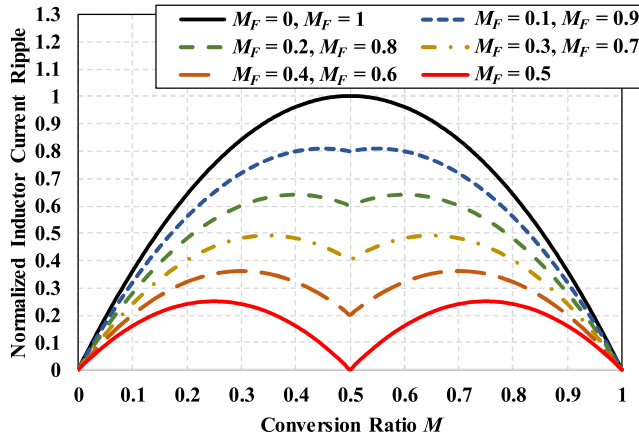


FIGURE 7. Normalized inductor current ripple of the 3-level buck converter according to flying capacitor voltage and output voltage.

where $D = D_A = D_B$, the volt-second balance of the inductor can be expressed as follows:

$$D(V_{in} - V_{Fly} - V_o) + D(V_{Fly} - V_o) = (1 - 2D)V_o. \quad (12)$$

In the high-gain region, where $D = D_A = D_B$, the volt-second balance of the inductor can be expressed as follows:

$$(1-D)(V_o - V_{Fly}) + (1-D)(V_o - V_{in} + V_{Fly}) = (2D - 1)(V_{in} - V_o). \quad (13)$$

From (12) and (13), the voltage conversion ratio $M = D$ can be satisfied. Thus, from (3)–(6), the normalized Δi_{Lo} according to M and M_F is illustrated in Fig. 7. As discussed previously, the inductor current ripple is the smallest at $M_F = 0.5$ and increases if M_F becomes larger or smaller than 0.5, as displayed in Fig. 7. Consequently, if V_{Fly} is not guaranteed to be $V_{in}/2$, the voltage stress on the switch will not be maintained at $V_{in}/2$ and there will be an imbalance in inductor current. This will result in increased current stress and circuit losses due to the higher current ripple, which is disadvantageous in terms of reducing the size of the reactive components. In the next section, we introduce the conventional current mode control method and the proposed control method to ensure $M_F = 0.5$.

III. CONTROL METHOD

A. CONVENTIONAL CURRENT MODE CONTROL METHOD

Conventional control methods, such as peak and valley current mode control (CMC), have been proposed to ensure a balanced inductor current and maintain the flying capacitor voltage at half of the input voltage, as depicted in Fig. 8. Among these methods, Peak-CMC (P-CMC) turns on the main switch according to a clock signal (CLK) that operates at twice the switching frequency. Moreover, it detects the peak current of the inductor on a cycle-by-cycle basis to turn off the main switch according to the High Limit signal generated by comparing the output voltage control signal (V_e) with the inductor current, as depicted in Fig. 8(b). Specifically, a clock signal (D_{CLK}) of the D Flip Flop is generated by the SR Flip Flop according to the CLK and High Limit signals.

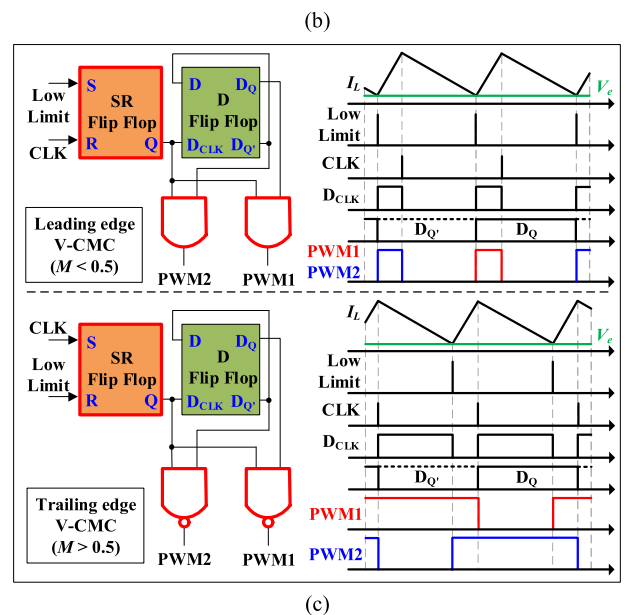
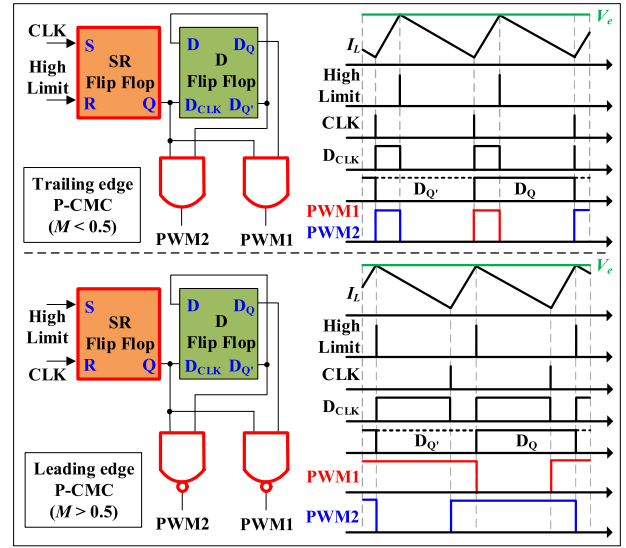
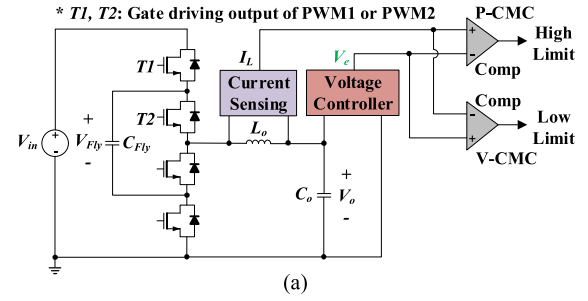


FIGURE 8. Block diagram of conventional CMC method: (a) 3-level buck converter circuit diagram with CMC method applied, (b) block diagram of P-CMC, and (c) block diagram of V-CMC.

The outputs D_Q and $D_{Q'}$ of the D Flip Flop then generate the main gate signals PWM1 and PWM2 through AND logic gates with D_{CLK} , respectively. This control mechanism maintains the peak current of the inductor at a constant level and ensures that the voltage across the flying capacitor is $V_{in}/2$. As depicted in Fig. 8(b), trailing-edge P-CMC can only

be applied in the low-gain region. If P-CMC is to be implemented in the high-gain region, the controller configuration must be changed to leading-edge P-CMC. This is because the turn-off signal of the main switch in the low-gain region should occur after the turn-on signal. In contrast, the turn-off signal of the main switch in the high-gain region should occur before the turn-on signal.

In Valley-CCM (V-CCM), the main switch is turned off according to a clock signal (CLK) that operates at twice the switching frequency. Here, the valley current of the inductor is detected instead of the peak current on a cycle-by-cycle basis to turn on the main switch according to the Low Limit signal generated by comparing the output voltage control signal (V_e) with the inductor current, as depicted in Fig. 8(c). Specifically, a clock signal (D_{CLK}) of the D Flip Flop is generated by the SR Flip Flop according to the CLK and Low Limit signals. The outputs D_Q and $D_{Q'}$ of the D Flip Flop then generate the main gate signals PWM1 and PWM2 through AND or NAND logic gates with D_{CLK} , respectively. This control mechanism maintains the valley current of the inductor at a constant level and ensures that the voltage across the flying capacitor is $V_{in}/2$. As depicted in Fig. 8(c), leading-edge V-CCM can only be applied in the low-gain region. If V-CCM is to be implemented in the high-gain region, the controller configuration must be changed to trailing-edge V-CCM. This is because the turn-on signal of the main switch in the low-gain region should occur before the turn-off signal. In contrast, the turn-on signal of the main switch in the high-gain region should occur after the turn-off signal [7]. Additionally, due to current subharmonic oscillations, both P-CMC and V-CCM require slope compensation to ensure normal operation in all input and output voltage operating ranges [18], [19], [20]. Moreover, as mentioned previously, CMC methods have complex control circuits (or algorithms) and are vulnerable to noise interference because they operate based on instantaneous peak or valley current sensing.

B. PROPOSED TWO INDEPENDENT SINGLE-LOOP VOLTAGE MODE CONTROL METHOD

The fundamental concept of the proposed control method is as follows: As discussed in Section II, in the low-gain region, when the duration $D_A T_s$ of operation in mode 1 is longer than the duration $D_B T_s$ of operation in mode 3, the charging energy of C_{Fly} increases, resulting in a higher flying capacitor voltage V_{Fly} . Furthermore, as the durations of $D_A T_s$ and $D_B T_s$ in modes 1 and 3 increase, the transferred energy to the output side also increases, resulting in a higher output voltage V_o . Similarly, in the high-gain region, when the duration $(1 - D_B) T_s$ of operation in mode 1 is longer than the duration $(1 - D_A) T_s$ of operation in mode 3, the charging energy of C_{Fly} increases, resulting in a higher flying capacitor voltage V_{Fly} . Moreover, if the duration $0.5(D_A + D_B - 1) T_s$ of operation in mode 4 increases, the energy transferred to the output side also increases, resulting in a higher output voltage V_o . In summary, V_{Fly} is proportional to D_A and V_o is proportional to both D_A and D_B . Therefore, D_A and D_B can be

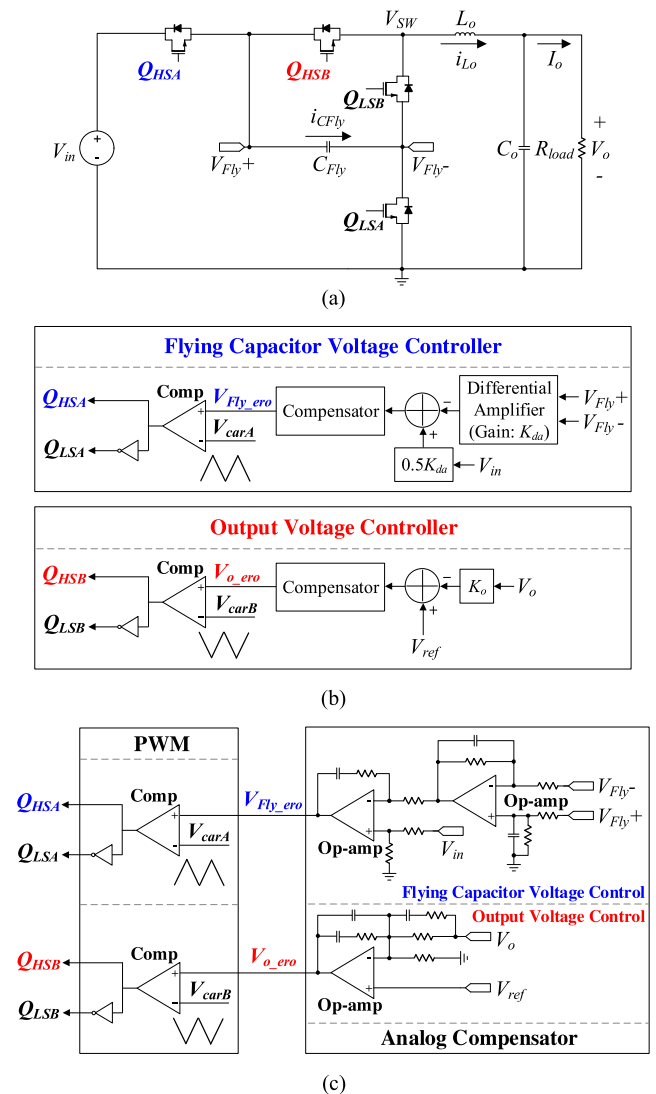


FIGURE 9. Block diagram of proposed two independent single-loop voltage mode control method: (a) 3-level buck converter circuit diagram, (b) simplified block diagram of proposed control method, and (c) proposed control method implemented using simple analog controllers.

employed to regulate V_{Fly} and V_o , respectively. Accordingly, the proposed control method can be implemented with two independent single-loop voltage mode controllers for V_{Fly} and V_o , as depicted in Fig. 9. As displayed in the simplified block diagram of the proposed control method in Fig. 9(b), V_o control is achieved through the pulse width modulation (PWM) gate signal. This is generated by comparing the control signal V_{o_ero} , which is generated after compensating for the error between sensed V_o and the control command V_{ref} , with the carrier signal V_{carB} . Independent from V_o control, V_{Fly} control is achieved through the PWM gate signal. This is generated by comparing the control signal V_{Fly_ero} , which is generated after compensating for the error between sensed V_{Fly} and the control command $V_{in}/2$, with the carrier signal V_{carA} , having a 180° phase difference with V_{carB} . Therefore, since the proposed control method is simple in terms of its

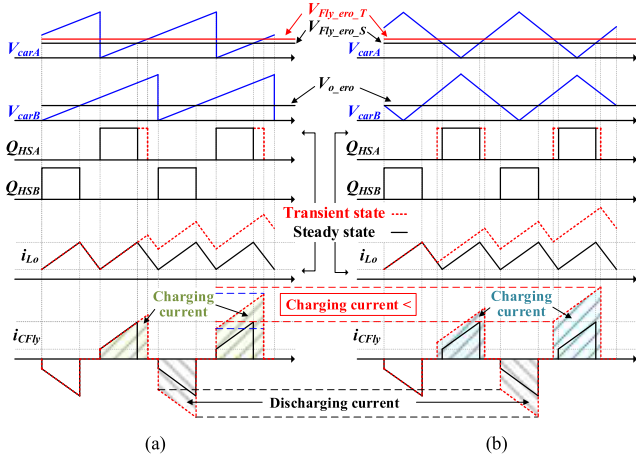


FIGURE 10. Differences in C_{Fly} charging and discharging currents according to the carrier signal during control operation (in the case of $V_{Fly} < V_{in}/2$, $M < 0.5$): (a) sawtooth wave carrier signal, (b) triangular wave carrier signal.

algorithm, it can be implemented with cost-effective and simple analog controllers, as depicted in Fig. 9(c). Moreover, the method is robust against noise because it only operates based on the output and flying capacitor voltages, without requiring information on the peak or valley currents. Additionally, since V_{Fly} and V_o are independently controlled, concerns about noise are mitigated, even if the sensing positions for V_{Fly} and V_o are spatially distant from each other, allowing for flexibility in component placement in PCB design. Furthermore, unlike conventional CMC methods, it ensures normal operation across all input-to-output voltage conversion ratios without changing the control configuration. The proposed control method exhibits different control dynamics depending on the waveform of the carrier signal. Specifically, compared to a sawtooth waveform, a triangular waveform can secure a larger charging and discharging current of C_{Fly} during transient states, resulting in faster and superior control dynamics. In the following section, we provide an example illustrating the transient characteristics according to the waveform of the carrier signal.

Fig. 10 and 11 illustrate examples of control operations depending on the carrier signal during transient states when V_{Fly} is lower than $V_{in}/2$ for $M < 0.5$ and $M > 0.5$, respectively. When $V_{Fly} < V_{in}/2$, the inductor current increases as D_A increases due to the increase in $V_{Fly_ero_S}$ to $V_{Fly_ero_T}$ caused by the flying capacitor voltage controller. In the low-gain region, the discharge current i_{CFly} of the flying capacitor remains constant, regardless of the carrier signal. In contrast, the charging current is larger when a triangular waveform carrier signal is applied. In the high-gain region, the charging current i_{CFly} remains constant regardless of the carrier signal. In contrast, the discharge current is smaller when a triangular waveform carrier signal is applied. Therefore, using the triangular waveform instead of the sawtooth waveform (even with the same duty cycle) results in a faster dynamic response and improved transient characteristics. When $V_{Fly} > V_{in}/2$, the

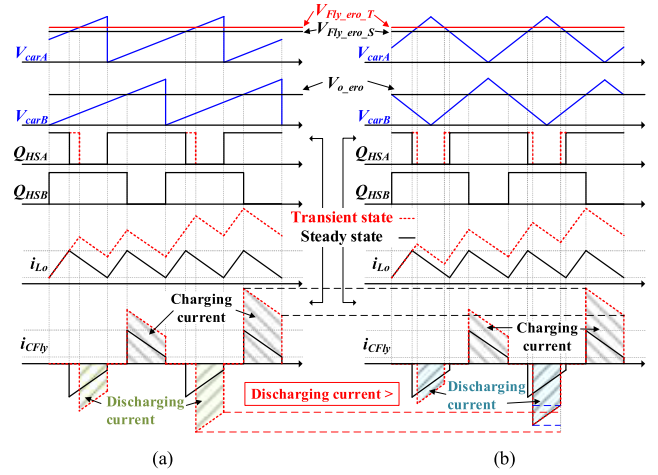


FIGURE 11. Differences in C_{Fly} charging and discharging currents according to the carrier signal during control operation (in the case of $V_{Fly} < V_{in}/2$, $M > 0.5$): (a) sawtooth wave carrier signal, (b) triangular wave carrier signal.

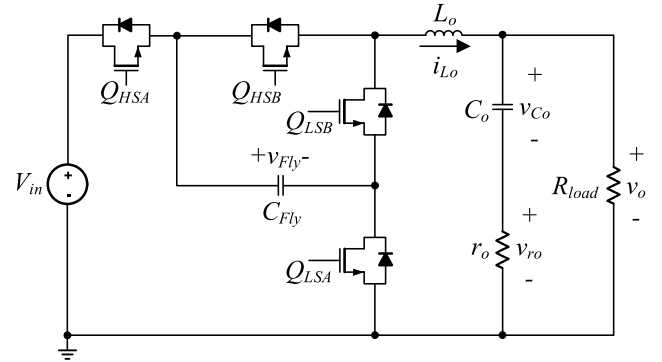


FIGURE 12. The 3-level buck converter with ESR of the output capacitor.

control operation based on the carrier signal is the opposite to that in Fig. 10 and 11.

C. MODELING OF THE 3-LEVEL BUCK CONVERTER WITH THE PROPOSED CONTROL METHOD

When modeling the 3-level buck converter, the traditional circuit-averaging method was adopted. Additionally, as illustrated in Fig. 12, the equivalent series resistance (ESR) r_o of the output capacitor is considered, influencing the dynamic characteristics. The 3-level buck converter has four operating modes, as displayed in Fig. 5. Next, we describe the voltage and current relationships for the output inductor, flying capacitor, and output capacitor in each operating mode.

In mode 1, where Q_{HSA} and Q_{LSB} switches turn on, the voltages and currents for each component are expressed as follows:

$$\begin{cases} L_o \frac{di_{Lo}}{dt} = V_{in} - v_{Fly} - v_o \\ C_{Fly} \frac{dv_{Fly}}{dt} = i_{Lo} \\ C_o \frac{dv_{Co}}{dt} = i_{Lo} - \frac{v_o}{R_{load}} = \frac{v_{ro}}{r_o} = \frac{v_o - v_{Co}}{r_o} \end{cases} \quad (14)$$

In mode 2, where Q_{LSA} and Q_{LSB} switches turn on, the voltages and currents for each component are expressed as

follows:

$$\begin{cases} L_o \frac{di_{L_o}}{dt} = -v_o \\ C_{Fly} \frac{dv_{Fly}}{dt} = 0 \\ C_o \frac{dv_{C_o}}{dt} = i_{L_o} - \frac{v_o}{R_{load}} = \frac{v_{r_o}}{r_o} = \frac{v_o - v_{C_o}}{r_o} \end{cases} \quad (15)$$

In mode 3, where Q_{LSA} and Q_{HSB} switches turn on, the voltages and currents for each component are expressed as follows:

$$\begin{cases} L_o \frac{di_{L_o}}{dt} = v_{Fly} - v_o \\ C_{Fly} \frac{dv_{Fly}}{dt} = -i_{L_o} \\ C_o \frac{dv_{C_o}}{dt} = i_{L_o} - \frac{v_o}{R_{load}} = \frac{v_{r_o}}{r_o} = \frac{v_o - v_{C_o}}{r_o} \end{cases} \quad (16)$$

In mode 4, where Q_{HSA} and Q_{HSB} switches turn on, the voltages and currents for each component are expressed as follows:

$$\begin{cases} L_o \frac{di_{L_o}}{dt} = V_{in} - v_o \\ C_{Fly} \frac{dv_{Fly}}{dt} = 0 \\ C_o \frac{dv_{C_o}}{dt} = i_{L_o} - \frac{v_o}{R_{load}} = \frac{v_{r_o}}{r_o} = \frac{v_o - v_{C_o}}{r_o} \end{cases} \quad (17)$$

In the low-gain region, the average variables for each component's voltage and current over one switching cycle, according to Fig. 4(a) and (14)–(16), can be expressed as follows:

$$\begin{cases} L_o \frac{d\langle i_{L_o} \rangle}{dt} = d_A V_{in} + (d_B - d_A) \langle v_{Fly} \rangle - \langle v_o \rangle \\ C_{Fly} \frac{d\langle v_{Fly} \rangle}{dt} = (d_A - d_B) \langle i_{L_o} \rangle \\ C_o \frac{d\langle v_{C_o} \rangle}{dt} = \langle i_{L_o} \rangle - \frac{\langle v_o \rangle}{R_{load}} = \frac{\langle v_{r_o} \rangle}{r_o} = \frac{\langle v_o \rangle - \langle v_{C_o} \rangle}{r_o} \end{cases} \quad (18)$$

where $\langle \cdot \rangle$ represents the average variable.

According to Fig. 4(b), in the high-gain region, the average variables for each component's voltage and current over one switching cycle can be expressed as derived from (14), (16), and (17), and are given by (18). Additionally, when the peak-to-peak amplitude of the carrier signal is 1, the variables d_A and d_B correspond to the control signals v_{Fly_ero} and v_{o_ero} , respectively. In this case, considering small signal perturbations, both control signals can be expressed as follows:

$$v_{Fly_ero} = V_{Fly_ero} + \Delta v_{Fly_ero}, \quad (19)$$

$$v_{o_ero} = V_{o_ero} + \Delta v_{o_ero}, \quad (20)$$

where V_{Fly_ero} and V_{o_ero} are dc values of the two control signals, and Δv_{Fly_ero} and Δv_{o_ero} denote the forced small-signal perturbation components of the two control signals. Furthermore, considering small-signal perturbations, the average state variable of the voltage and current for each component can be expressed as the sum of a dc value and a small-signal

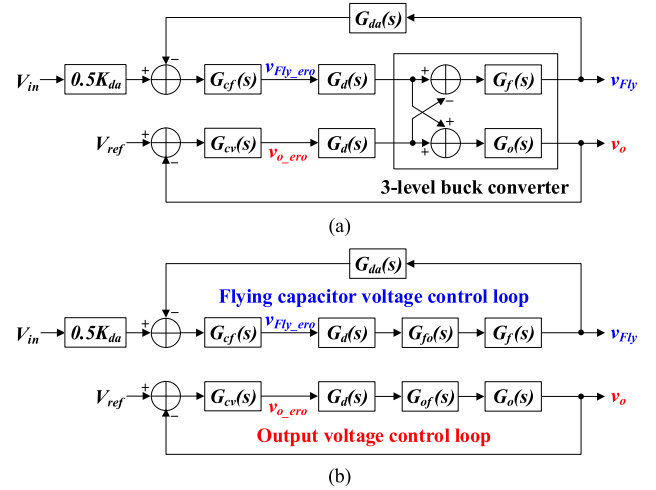


FIGURE 13. Control block diagram of the 3-level buck converter with proposed control method: (a) original system, (b) simplified system.

component, as follows: $\langle i_{L_o} \rangle = I_{L_o} + \Delta i_{L_o}$, $\langle v_{Fly} \rangle = V_{Fly} + \Delta v_{Fly}$, $\langle v_o \rangle = V_o + \Delta v_o$, $\langle v_{C_o} \rangle = V_{C_o} + \Delta v_{C_o}$, $\langle v_{r_o} \rangle = V_{r_o} + \Delta v_{r_o}$. Substituting (19), (20), and the average state variable into (18) and extracting the first-order small-signal components, the following can be derived:

$$\begin{cases} L_o \frac{d\Delta i_{L_o}}{dt} = 0.5V_{in} (\Delta v_{Fly_ero} + \Delta v_{o_ero}) - \Delta v_o \\ C_{Fly} \frac{d\Delta v_{Fly}}{dt} = I_{L_o} (\Delta v_{Fly_ero} - \Delta v_{o_ero}) \\ C_o \frac{d\Delta v_{C_o}}{dt} = \Delta i_{L_o} - \frac{\Delta v_o}{R_{load}} = \frac{\Delta v_{r_o}}{r_o} = \frac{\Delta v_o - \Delta v_{C_o}}{r_o} \end{cases} \quad (21)$$

The 3-level buck converter with the proposed control method is a system that comprises two input perturbations (Δv_{Fly_ero} and Δv_{o_ero}) and two output perturbations (Δv_{Fly} and Δv_o).

By transforming (21) into the Laplace domain, the input and output perturbation relationship can be derived as follows:

$$\Delta v_o = G_o(s) (\Delta v_{Fly_ero} + \Delta v_{o_ero}), \quad (22)$$

$$\Delta v_{Fly} = G_f(s) (\Delta v_{Fly_ero} - \Delta v_{o_ero}), \quad (23)$$

where $G_o(s)$ and $G_f(s)$ are given by

$$G_o(s) = \frac{\frac{V_{in} r_o}{2L_o} \left(s + \frac{1}{r_o C_o} \right)}{\left(1 + \frac{r_o}{R_{load}} \right) s^2 + \left(\frac{r_o}{L_o} + \frac{1}{R_{load} C_o} \right) s + \frac{1}{L_o C_o}}, \quad (24)$$

$$G_f(s) = \frac{I_{L_o}}{s C_{Fly}}. \quad (25)$$

Accordingly, the overall system control block diagram in Fig. 9 can be equivalently represented as depicted in Fig. 13(a). Here, $G_{cf}(s)$ and $G_{cv}(s)$ are the transfer functions of the flying capacitor voltage control compensator and the output voltage control compensator, respectively. $G_{da}(s)$ is the transfer function of the differential amplifier for flying capacitor voltage sensing, as displayed in Fig. 9. If a micro-controller is employed for PWM implementation (rather than

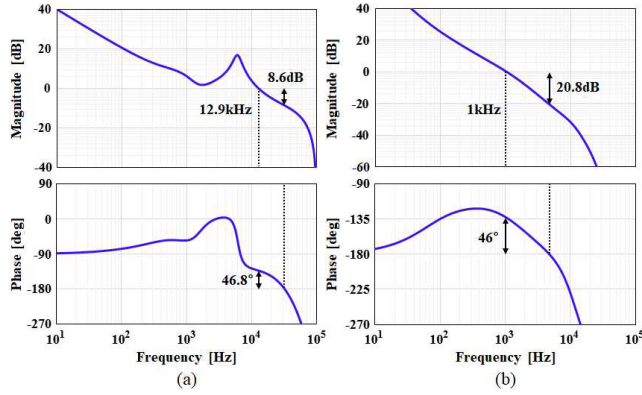


FIGURE 14. Bode plots: (a) output voltage control loop, (b) flying capacitor voltage control loop.

an analog PWM), the control loop should incorporate the transfer function $G_d(s)$, which represents the computation delay, zero-order holder, and modulation gain, as depicted in Fig. 13. However, because the two control loops presented in Fig. 13(a) influence each other, designing the controllers can be complex.

Therefore, to facilitate controller design, the circuit model of the 3-level buck converter and the two control loops have been modeled together. This process is outlined as follows:

Considering small-signal perturbations, the perturbed input and output signal relationship of the 3-level buck converter in Fig. 13(a) can be expressed as follows:

$$\Delta v_o = G_o(s) G_d(s) (\Delta v_{Fly_ero} + \Delta v_{o_ero}), \quad (26)$$

$$\Delta v_{Fly} = G_f(s) G_d(s) (\Delta v_{Fly_ero} - \Delta v_{o_ero}), \quad (27)$$

$$\Delta v_{Fly_ero} = -G_{da}(s) G_{cf}(s) \Delta v_{Fly}, \quad (28)$$

$$\Delta v_{o_ero} = -G_{cv}(s) \Delta v_o. \quad (29)$$

From (26)–(29), the transfer function $G_{oo}(s)$ for the output voltage control signal Δv_{o_ero} to the output voltage Δv_o and $G_{ff}(s)$ for the flying capacitor voltage control signal Δv_{Fly_ero} to the flying capacitor voltage Δv_{Fly} are as follows:

$$G_{oo}(s) = \left\{ \frac{1 + 2G_{da}(s) G_{cf}(s) G_d(s) G_f(s)}{1 + G_{da}(s) G_{cf}(s) G_d(s) G_f(s)} \right\} G_o(s) G_d(s) = G_{of}(s) G_o(s) G_d(s), \quad (30)$$

$$G_{ff}(s) = \left\{ \frac{1 + 2G_{cv}(s) G_d(s) G_o(s)}{1 + G_{cv}(s) G_d(s) G_o(s)} \right\} G_f(s) G_d(s) = G_{fo}(s) G_f(s) G_d(s). \quad (31)$$

Accordingly, the overall system control block diagram in Fig. 13(a) can be simplified as depicted in Fig. 13(b), with the flying capacitor voltage control loop and the output voltage control loop displayed as independent configurations.

With specifications of $V_{in} = 66$ V, $V_o = 32$ V, $R_{load} = 2.048 \Omega$, $f_{sw} = 100$ kHz, $L_o = 1.36$ uH, $C_{Fly} = 20$ uF, $C_o = 500$ uF, and $r_o = 13$ mΩ, the Bode plots for the output voltage control loop and the flying capacitor voltage control loop are displayed in Fig. 14(a) and (b), respectively. In the prototype design, the crossover frequency for the output voltage control loop was selected as 12.9 kHz with a phase margin of 46.8°

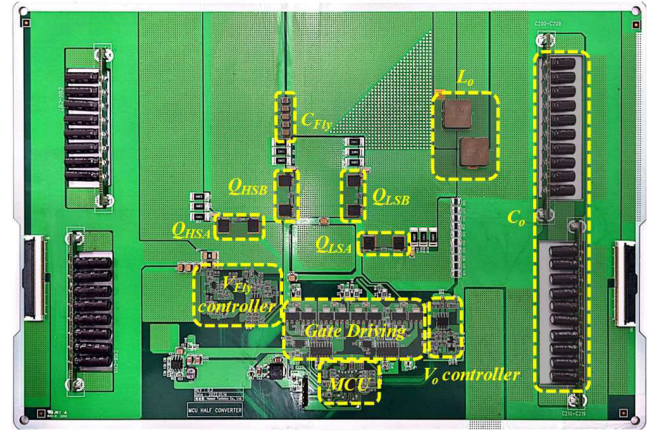


FIGURE 15. 500 W-rated prototype of a 3-level buck converter.

TABLE 2. Applied parameters in the experiment.

Specifications		
Input voltage V_{in}	48–70 V / Nominal 66 V	
Output voltage V_o	32 V	
Rated power P_o	Maximum 500 W	
Switching frequency f_{sw}	100 kHz	
Components		
Switch	Q_{HSA}, Q_{LSA}	BSC034N10LS5 * 2Parallel
	Q_{HSB}, Q_{LSB}	BSC027N06LS5 * 2Parallel
Output inductor L_o	IHLP-5050CE-01 * 2 Series (Thickness 3.5 mm)	
	0.68 uH * 2 in series (Total 1.36 uH)	
Microcontroller	dsPIC33FJ09GS302	

and a gain margin of 8.6 dB. The crossover frequency for the flying capacitor voltage control loop was selected as 1 kHz with a phase margin of 46° and a gain margin of 20.8 dB.

IV. EXPERIMENTAL RESULTS

To validate the feasibility of the proposed control method, we present experimental results obtained from a 500 W-rated prototype of the 3-level buck converter depicted in Fig. 15. The proposed control method was implemented with simple analog controllers and incorporated digital PWM implemented on the dsPIC33FJ09GS302, as displayed in Fig. 9(c). The specifications and main component information for the experiment are detailed in Table 2. Given that V_{Fly} was approximately 0 V during the startup transients, V_{in} was applied to Q_{HSA} and Q_{LSA} following the operation in modes 1 and 3. Therefore, we needed to consider the maximum voltage stresses on Q_{HSA} and Q_{LSA} given the maximum input voltage of 70 V. For Q_{HSB} and Q_{LSB} , we considered a maximum steady-state operating voltage of 35 V because V_{Fly} was applied. Hence, we selected switches with different rated voltages for each. By using an output inductor with a thickness of 3.5 mm, we achieved an overall prototype thickness of 6 mm, which was suitable for ultra-slim applications.

Fig. 16 displays the experimental steady-state waveforms according to the input voltage V_{in} at full load, including

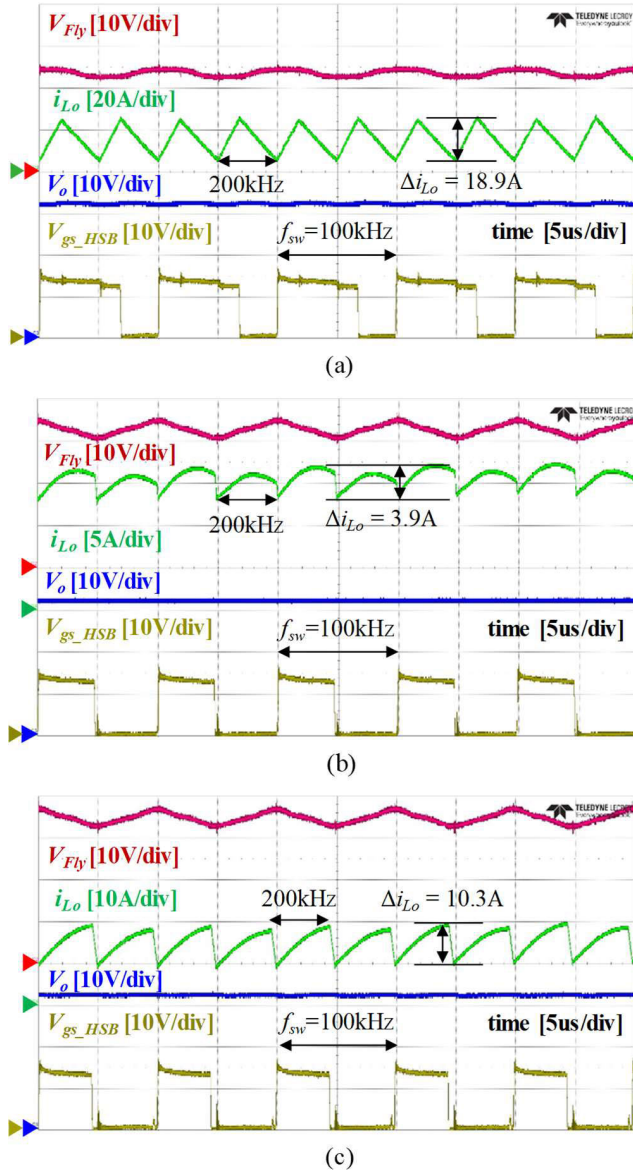


FIGURE 16. Experimental steady-state waveforms according to V_{in} at full load: (a) $V_{in} = 48$ V, (b) $V_{in} = 66$ V, and (c) $V_{in} = 70$ V.

the flying capacitor voltage V_{Fly} , the output inductor current i_{Lo} , the output voltage V_o , and the gate signal of Q_{HSB} (V_{gs_HSB}). Fig. 16(a) and (c) display the cases when V_{in} is at the minimum value of 48 V ($M > 0.5$) and the maximum value of 70 V ($M < 0.5$), respectively. We achieved inductor current balance for the switching half-cycle by maintaining the average V_{Fly} at $V_{in}/2$, which is 24 V in Fig. 16(a) and 35 V in Fig. 16(c). Accordingly, by maintaining the average V_{Fly} at $V_{in}/2$ over the entire input voltage range, we achieved inductor current balance for the switching half-cycle and controlled V_o at 32 V in a steady state, validating the feasibility of the proposed control method. When V_{in} was at the nominal value of 66 V ($M \approx 0.5$), the inductor current ripple ideally became zero, as illustrated in Fig. 2. However, as depicted in Fig. 16(b), the inductor current followed a resonant waveform due to the resonance between C_{Fly} and L_o in modes 1 and 3,

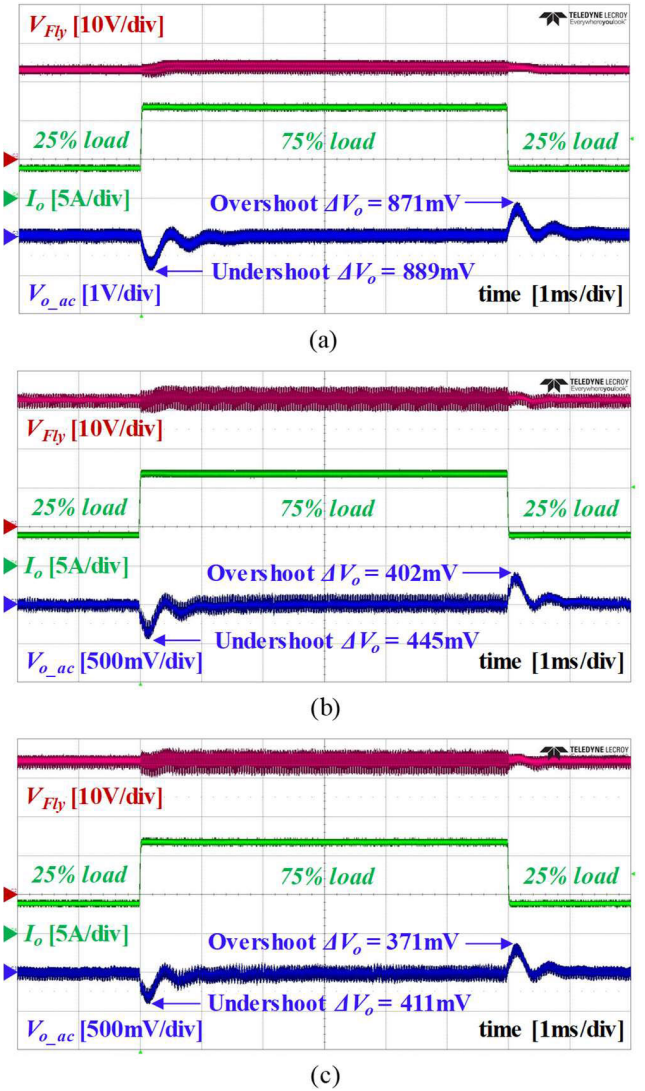


FIGURE 17. Experimental transient response waveform according to V_{in} as the load changes from 25% to 75%, and vice versa: (a) $V_{in} = 48$ V, (b) $V_{in} = 66$ V, and (c) $V_{in} = 70$ V.

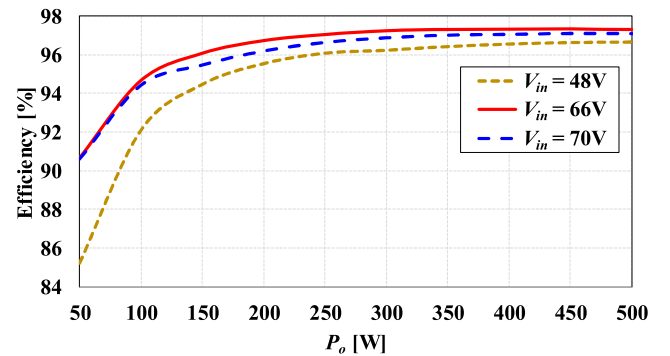


FIGURE 18. Measured efficiencies of a 3-level buck converter.

resulting in practical inductor current ripple. Additionally, the current ripple increased slightly due to a small current imbalance, although it remained sufficiently small to be considered negligible.

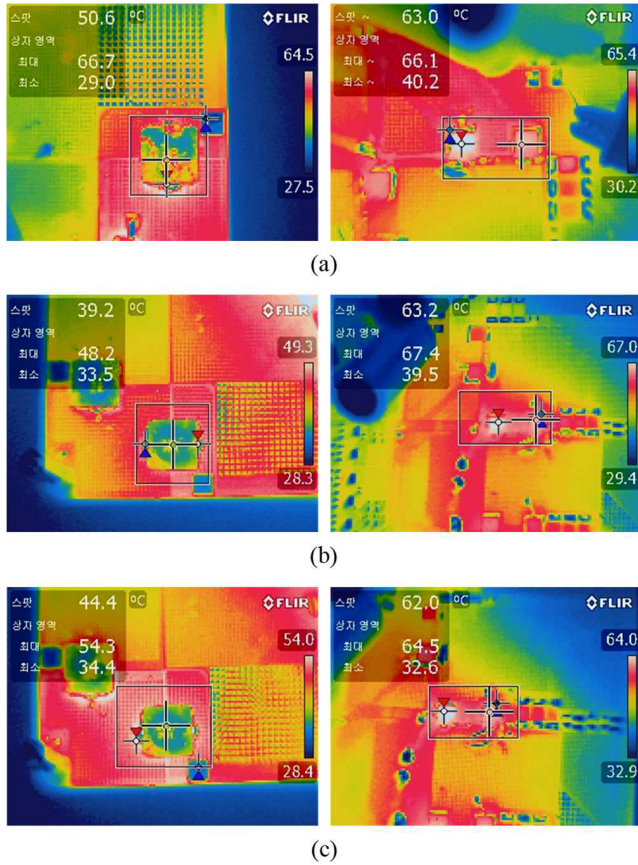


FIGURE 19. Thermal performances of key components according to V_{in} at full load (ambient temperature: 28°C). Left images: inductor, right images: main switch with the highest recorded temperatures. (a) $V_{in} = 48$ V, (b) $V_{in} = 66$ V, and (c) $V_{in} = 70$ V.

Fig. 17 displays the experimental transient response wave-form according to V_{in} as the load changed from 25% to 75%, and *vice versa*. The output voltage was effectively regulated within a 3% range across the entire input voltage range during the load variations. At the minimum input voltage of 48 V in Fig. 17(a), the output voltage variation was the greatest, with an overshoot of 871 mV and an undershoot of 889 mV. The average flying capacitor voltage was successfully maintained at $V_{in}/2$. These results confirmed that the proposed control method could effectively regulate both the output voltage and the flying capacitor voltage, indicating good load transient performance.

Fig. 18 displays the measured efficiencies for various loads according to the input voltage. As the voltage conversion ratio M became larger or smaller than 0.5, the inductor current ripple increased, as illustrated in Fig. 2, resulting in higher conduction losses. As displayed in Fig. 16, the inductor current ripple was 18.9, 3.9, and 10.3 A for input voltages of 48, 66, and 70 V, respectively. Therefore, the efficiency was lowest at $V_{in} = 48$ V, where the inductor current ripple was largest. In contrast, the efficiency was highest at $V_{in} = 66$ V, where the current ripple was smallest, achieving a maximum efficiency of 97.35% at $P_o = 450$ W.

Fig. 19 displays the thermal performances of key components according to V_{in} at full load, with measurements taken at an ambient temperature of 28°C . The left and right images depict the thermal performances of the inductor and main switch, respectively, highlighting the components with the highest recorded temperatures. Overall, the prototype maintained a respectable temperature in the 60°C range, even without a heatsink. The inductor's heat generation was highest at 66.7°C when $V_{in} = 48$ V, where the inductor current ripple was largest. In contrast, the inductor's heat generation was lowest at 48.2°C when $V_{in} = 66$ V, where the inductor current ripple was smallest. These results indicated that the efficiency measurements presented in Fig. 18 are reliable.

V. CONCLUSION

In this paper, we discussed the 3-level buck converter, which offers significant advantages in terms of reducing the size of reactive components that occupy a large volume. To ensure the current balance of L_o for every switching half cycle, V_{Fly} of the 3-level buck converter must be maintained at $V_{in}/2$. To achieve current balance by maintaining V_{Fly} at $V_{in}/2$ across all input-to-output voltage operating ranges while simultaneously controlling V_o , we proposed a two independent single-loop voltage mode control method. This method consisted of one controller to maintain V_{Fly} at $V_{in}/2$ and another to regulate V_o .

Conventional CMC methods rely on instantaneous peak or valley current sensing, are susceptible to noise interference, and have complex control circuits or algorithms. Furthermore, when both P-CMC and V-CMC are needed for $M < 0.5$ and $M > 0.5$, different control configurations must be employed based on M (a leading-edge current control scheme and a trailing-edge current control scheme), as depicted in Fig. 8.

In contrast, the proposed control method is robust to noise interference because it operates solely based on V_{Fly} and V_o . Moreover, it can also be implemented simply using low-cost digital or analog controllers without the need for current sensing. Additionally, the proposed approach ensures normal operation across all M values without changing the control configuration.

An experimental validation of the proposed control method was conducted using a 500 W-rated prototype, where a high efficiency of 97.35% was achieved at an input voltage of 66 V. Moreover, the 3-level buck converter significantly reduced the output inductor current ripple compared to conventional buck converters. This reduction allowed the use of very small inductors with a thickness of 3.5 mm, ultimately resulting in an ultra-slim design for the entire prototype, with a thickness of 6 mm.

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